



Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere-04
 An Autonomous Institute Affiliated to VTU
 Department of CSE/ISE & Allied Branches



Course Title	Digital Design & Computer Organization (DD&CO)	Course Code	BCSPCC303
Course Instructors	Prof. Vijayalakshmi S Hallikeri, Prof. Divya Patil, Prof. Parveen Taj, Prof. Archana T C, Prof. Rahima, Prof. Sanjay	Semester-Section	3 *
CIE Test	1 st	Max. Marks	30
Date	15.11.2025	Time	9:30am-10:45am

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Apply the K-Map techniques to simplify various Boolean expressions.
CO2	Design different types of combinational and sequential circuits along with Verilog programs.
CO3	Describe the fundamentals of machine instructions, addressing modes and processor performance.
CO4	Explain the approaches involved in achieving communication between processor and I/O devices.
CO5	Analyse internal organization of memory and impact of cache/pipelining on processor performance.

Note: Answer any ONE full question from each Part.

Q. No.		Question	Marks	RBT Level	CO
Part-A					
1	A	State and explain Duality principle & De-morgan's theorem with an example.	6	L2	1
	B	Simplify the following Boolean function using K-map and Implement using i. NAND Gates only ii. NOR Gates only $F(A,B,C,D) = \sum m(1,5,6,12,13,14) + (2, 4)$	6	L3	1
OR					
2	A	Simplify the expression $F = (x+y)(x^1+z)(y+z)$ algebraically.	6	L3	1
	B	Simplify the following Boolean function using Quine-McClusky method: $F(A,B,C,D) = \sum m(0,2,3,6,7,) + (5,8,10,11)$	6	L3	1
Part-B					
3	A	Design and implement full-adder with (i) AND-OR (ii) XOR gates and write the HDL Verilog code for full-adder.	6	L3	2
	B	Define the following terms in relation to flip-flops: (i) Setup time (ii) Hold time (iii) Propagation delay time.	6	L2	2
OR					
4	A	Implement $F(A,B,C,D) = A^1BD + A^1C^1D + B^1CD$ using 8:1 MUX.	6	L3	2
	B	What is flip-flop? Explain the working of any two: (i) S-R flip-flop (ii) D flip-flop (iii) J-K flip-flop (iv) T flip-flop	6	L2	2
Part-C					
5		Describe basic functional units of a computer with a neat block diagram.	6	L2	3
OR					
6		Explain (i) Processor Clock (ii) Clock Rate (iii) Basic Performance Equation	6	L2	3

RBT (Revised Bloom's Taxonomy) Levels: Cognitive Domain		
L1: Remembering	L2: Understanding	L3: Applying
L4: Analysing	L5: Evaluating	L6: Creating

Course Instructor / Coordinator

Coordinator DAAC

Program Coordinator

BoE Chairman



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Bapuji Institute of Engineering and Technology, Davanagere-04
 An Autonomous Institute Affiliated to VTU
 Department of Artificial Intelligence & Machine Learning (AI&ML)

Course Title	Digital Design & Computer Organization (DD&CO)	Course Code	BCSPCC303
Course Instructors	Prof. Vijayalakshmi S Hallikeri, Prof. Divya Patil, Prof. Parveen Taj, Prof. Archana T C, Prof. Rahima B, Prof. Sanjay	Semester-Section	3 *
CIE Test	2 nd	Max. Marks	30
Date	09.01.2026	Time	9:30am-10:45am

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Apply the K-Map techniques to simplify various Boolean expressions.
CO2	Design different types of combinational and sequential circuits along with Verilog programs.
CO3	Describe the fundamentals of machine instructions, addressing modes and processor performance.
CO4	Explain the approaches involved in achieving communication between processor and I/O devices.
CO5	Analyse internal organization of memory and impact of cache/pipelining on processor performance.

Note: Answer any ONE full question from each Part.					
Q. No.		Question	Marks	RBT Level	CO
Part-A					
1	a	Discuss the following: (i) Byte addressability (ii) Big endian (iii) Little endian assignments.	6	L2	3
	b	Define addressing mode. Explain the different addressing modes with an example for each.	6	L2	3
OR					
2	a	Write one address, two address and three address instructions to carry out: $C \leftarrow A+B$.	6	L3	3
	b	Demonstrate the Straight-line sequencing or Branching operations using a loop to add n numbers with neat diagrams.	6	L3	3
Part-B					
3	a	What is an interrupt? Discuss various methods of handling simultaneous interrupt requests from multiple devices.	6	L2	4
	b	What is bus arbitration? Describe the different types of bus arbitration.	6	L2	4
OR					
4	a	Demonstrate DMA and its implementation and show how data is transferred between memory and I/O devices using DMA controller.	6	L2	4
	b	What is cache memory? Explain the different mapping functions used in cache memory.	6	L2	4
Part-C					
5		What is pipelining? Explain 4-stage pipelining with neat diagram.	6	L2	5
OR					
6		Write and discuss the control sequence for execution of the instruction: Add (R3), R1	6	L3	5

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UBH
 Course Coordinator

Naven
 Coordinator DQAC

Mjw
 Program Coordinator
 (HOD, AI&ML)

Cp
 BoE Chairman